

It's *GO TIME*!

Quantifying Tag-Up Reaction Time to Increase Fan Engagement

SMT Data Challenge 2026

Team 204

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Abstract

The outcome of a baseball game is often decided by inches. When a runner on third base tags up, beating the throw home requires more than raw speed, it requires a split-second reaction. In this paper, we uncover the hidden race before the sprint, translating a runner's reaction time into a dynamic new metric: *GO TIME*. Using minor league player tracking data, we analyzed 109 competitive tag-up plays to measure the exact delay between the outfielder's catch and the runner's break for home. To make it even more engaging, we developed stadium scoreboard graphics to display a player's *GO TIME* and how his *GO TIME* ranks. The scoreboard graphic also puts players' times into perspective, using comparisons such as the time it takes to blink, or how fast a fastball reaches the plate. We also created an interactive fan experience where fans can test their own *GO TIMES* against other fans and baseball's best. *GO TIME* leverages advanced player tracking along with an immersive fan experience to offer a deeper insight into baserunning efficiency while also boosting fan engagement.